

InfoMorph

Project Team

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Chapter 1

Introduction

This document is to serve as the scope document of our Final Year Project(FYP). Our Project InfoMorph revolutionizes the concept of consuming Information by providing concise and on-topic summarization of textual data and generating an AI video narrating the topic.

The world of today has witnessed a significant transition away from traditional news sources and toward social media. Research shows that individuals between the ages of 12 and 24 absorbed 76% of their news on social media platforms[1]. As a result, young people do not receive accurate information and become unaware of their environment. Additionally, it was observed that 83% of individuals prefer to learn information visually[3]. The social media era and changing times have altered how individuals live their lives and get their information.

With all of these insights in mind, we developed the concept of InfoMorph: Revolutionizing Information with AI Narration, which will use the aforementioned techniques to address these problems.

1.1 Problem Statement

Today's youth, especially those between the ages of 12 and 24, have difficulties digesting information in the fast-paced digital world. Many people feel that reading articles takes up time, and they would much rather engage in other activities. In addition, there is an excess of information available in the current internet era, a phenomenon referred to as information overload. People become confused about what to read and watch as a result, missing out on any information at all. Furthermore, articles with no real substance or useful information are being published just for eye-catching headlines due to the clickbait culture. As a result, young people are becoming less interested in keeping up with current affairs, which has left them ignorant of the world around them. Addressing these issues

requires a solution that provides relevant, digestible, and visually engaging content.

1.2 Scope

Our project aims to transform how we consume information from the internet into something quicker, more engaging, and more informative. We want to create a dynamic project that gathers information from articles, websites, and other platforms, converting it all into a single summary that is to the point. This summary will then be converted from text to speech, and that will be done using an open source model, which will support both English and Urdu. Moreover, our custom AI anchor will deliver the news, which we will train on our own dataset using the First Order Motion Model (FOMM). Our project's main aim is to keep the content we gather concise so it is easier for our users to stay up to date with the information they require with them needing to read lengthy articles or websites. The videos are set to have a length between 30 to 120 seconds, as we feel this is the perfect duration for keeping our users interested while still conveying all the main points.

1.3 Modules

1.3.1 Module 1

UI/UX design: We will create an interactive design for our project where we will try to focus mainly on topic selection so our users can engage with the information.

1.3.2 Module 2

Web Scraping: Here we will focus on automating the data extraction of all the articles, websites and other online sources that have data related to the content that the user requires.

1.3.3 Module 3

Summarization: Using NLP techniques and a LLM, we will convert all the information we have gathered into a single summary that is concise and to the point.

1.3.4 Module 4

Text-To-Speech: The summary from the previous module will be converted into audio using an open source model for our AI News Anchor to say.

1.3.5 Module 5

Video Generation: This module will focus on the generation of video which will show our AI News Anchor and it will be done using First Order Motion Model where we will train our own news anchor and what he/she will do.

1.3.6 Module 6

Integration: All the previous modules will be connected here to make sure that the data flow is smooth and that all individual modules communicate with each other correctly.

1.4 User Classes and Characteristics

User class	Description
General Users	Individuals seeking to get quick information about a certain topic Goals: Quickly consume news in a visual and entertaining way. Frequency of Use: Daily or occasional users who scroll through social media feeds. Skill Level: Basic technical proficiency (Be able to write the query for the topic). View: User View
Platform Administrators	Admins responsible for managing the InfoMorph platform, ensuring smooth operation, handling user issues, and maintaining content quality. Goals: Ensure platform functionality, user satisfaction, and video content quality. Frequency of Use: Daily monitoring and managing of the platform. Skill Level: Advanced technical proficiency (familiar with back-end systems, troubleshooting, user support).

Example: User Classes and Characteristics

Chapter 2

Project Requirements

In this chapter, we will tell you about InfoMorph's functional and non-functional requirements as well as use cases.

2.1 Use-case Diagram

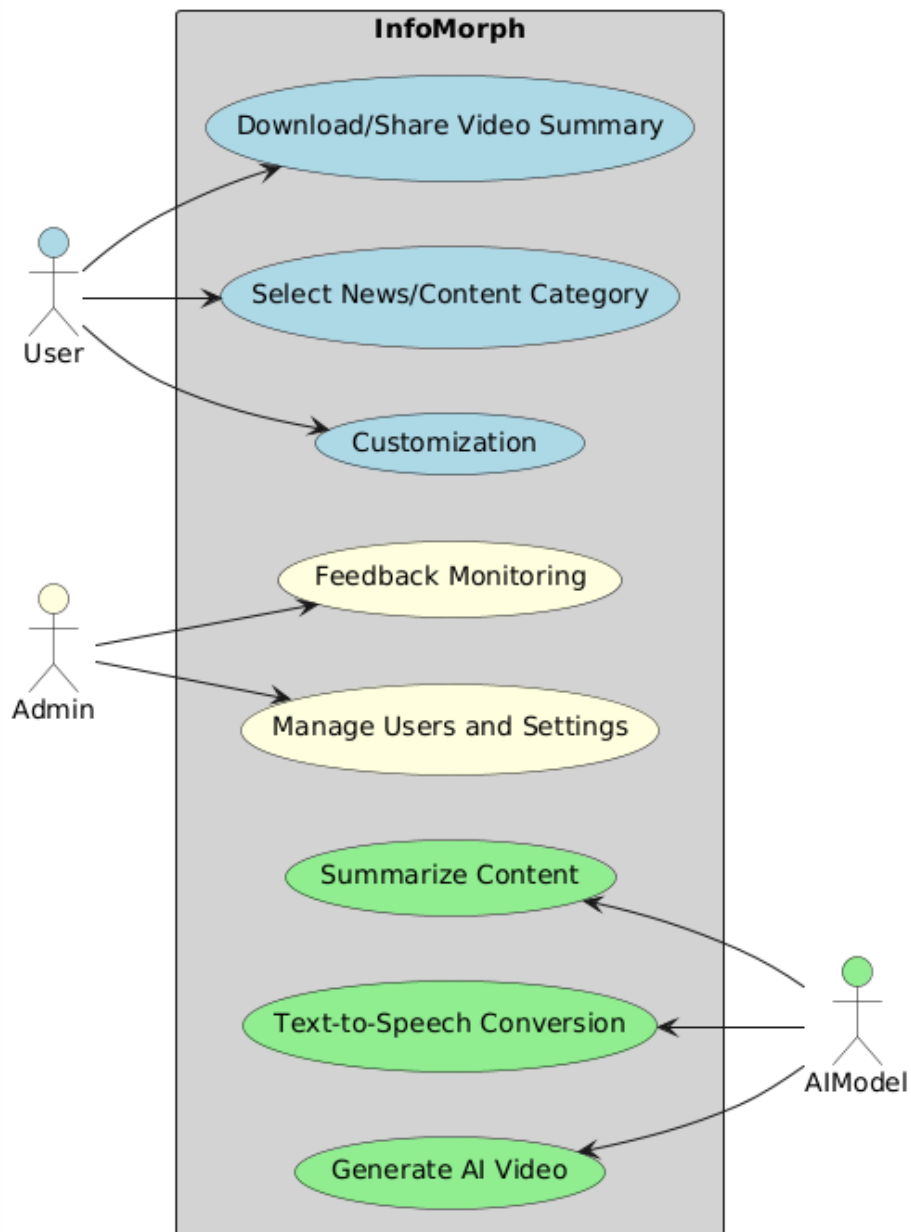


Figure 2.1: Use Case Diagram

2.2 High-Level Use Cases

2.2.1 Use Case 1

Name	Select Topic
Actors	User
Type	Primary Actor
Description	The user can select the topic news, and also the user can select the input and output language . Upon selecting a topic, the system will display relevant URLs.

2.2.2 Use Case 2

Name	Summarize Content
Actors	AI Model
Type	Secondary Actor
Description	The user provides a piece of content that they want summarized. The system processes the content and returns a concise summary using its AI summarization capabilities.

2.2.3 Use Case 3

Name	Text-to-Speech Conversion
Actors	AI model
Type	Secondary Actor
Description	The summarized content is converted into speech. The system generates an audio file of the summary, allowing the user to listen to it.

2.2.4 Use Case 4

Name	Generate AI Video
Actors	AI Model
Type	System Actor
Description	The system takes the audio file generated from the text-to-speech conversion and produces a video featuring an AI anchor. This video can then be accessed by the user for a more engaging experience.

2.2.5 Use Case 5

Name	Download/Share Video Summary
Actors	User
Type	Primary Actor
Description	The user selects a video summary that they wish to download or share. The system allows the user to download the video file or share it through various platforms.

2.2.6 Use Case 6

Name	Customization
Actors	User
Type	Primary Actor
Description	The user can customize settings related to content delivery and notifications. The system saves these preferences and applies them to enhance the user's experience.

2.2.7 Use Case 7

Name	Feedback Monitoring
Actors	Admin
Type	Secondary Actor
Description	The admin can monitor and analyze user feedback on the content and services provided. This information helps improve the system and address any issues raised by users.

2.2.8 Use Case 8

Name	Manage Users and Settings
Actors	Admin
Type	Secondary Actor
Description	The admin has the ability to manage user accounts and their associated settings. This includes editing user details, activating or deactivating accounts, and ensuring proper user management.

2.3 Functional Requirements

2.3.1 Module 1

Following are the requirements for module 1:

1. Create a simple User interface that will allow users to navigate and operate our application easily
2. Users will be able to select whatever topic they want information regarding.

2.3.2 Module 2

Following are the requirements for module 2:

1. Create a web scraping system that can automatically gather articles and data from specified internet URLs.

2.3.3 Module 3

Following are the requirements for module 3:

1. We will be using NLP methods and a LLM to examine all the data collected from Web scraping and converting it into a brief and reliable summary
2. We will ensure the summary contains what was in the scrapped articles and none of the important points are lost.

2.3.4 Module 4

Following are the requirements for module 4:

1. We will use Text-To-Speech (TTS) systems that are already available to convert the summarized text into speech.

2.3.5 Module 5

Following are the requirements for module 5:

1. We will use FOMM(First Order Motion Model) to generate an AI News Anchor that will deliver the news with real facial expressions
2. We will train the model on different gestures with FOMM and give a range of anchors for customized news presentations.

2.4 Non-Functional Requirements

2.4.1 Reliability

- The system shall maintain a 99.5% uptime to ensure consistent access to news and articles.
- In the event of a failure, the system shall recover and resume service within 10 minutes.
- Error logs shall be generated for every failure and stored for debugging purposes.

2.4.2 Usability

- The user interface shall allow users to select and categorize news or articles with no more than 4 clicks.
-

2.4.3 Performance

- 90% of summaries shall be generated within 20 seconds for articles of up to 1000 words.
- The system shall be able to handle at least 100 concurrent users without degradation in performance.
- AI-generated videos of article summaries shall be produced and made available within 2 minutes of the request.

2.4.4 Security

- The system shall implement encryption (AES-256) for all user data to ensure data privacy.

- Users shall be required to authenticate using multi-factor authentication (MFA) to prevent unauthorized access.

2.4.5 Maintainability

- The codebase shall be modular to allow for easy updates and bug fixes without impacting overall system performance.
- Documentation shall be maintained to ensure that new developers can understand the system within 2 weeks.

2.4.6 Scalability

- The system architecture shall support horizontal scaling to handle an increased number of users and data volume.
- The platform shall be able to scale and process up to 1000 articles per hour.

2.5 Domain Model

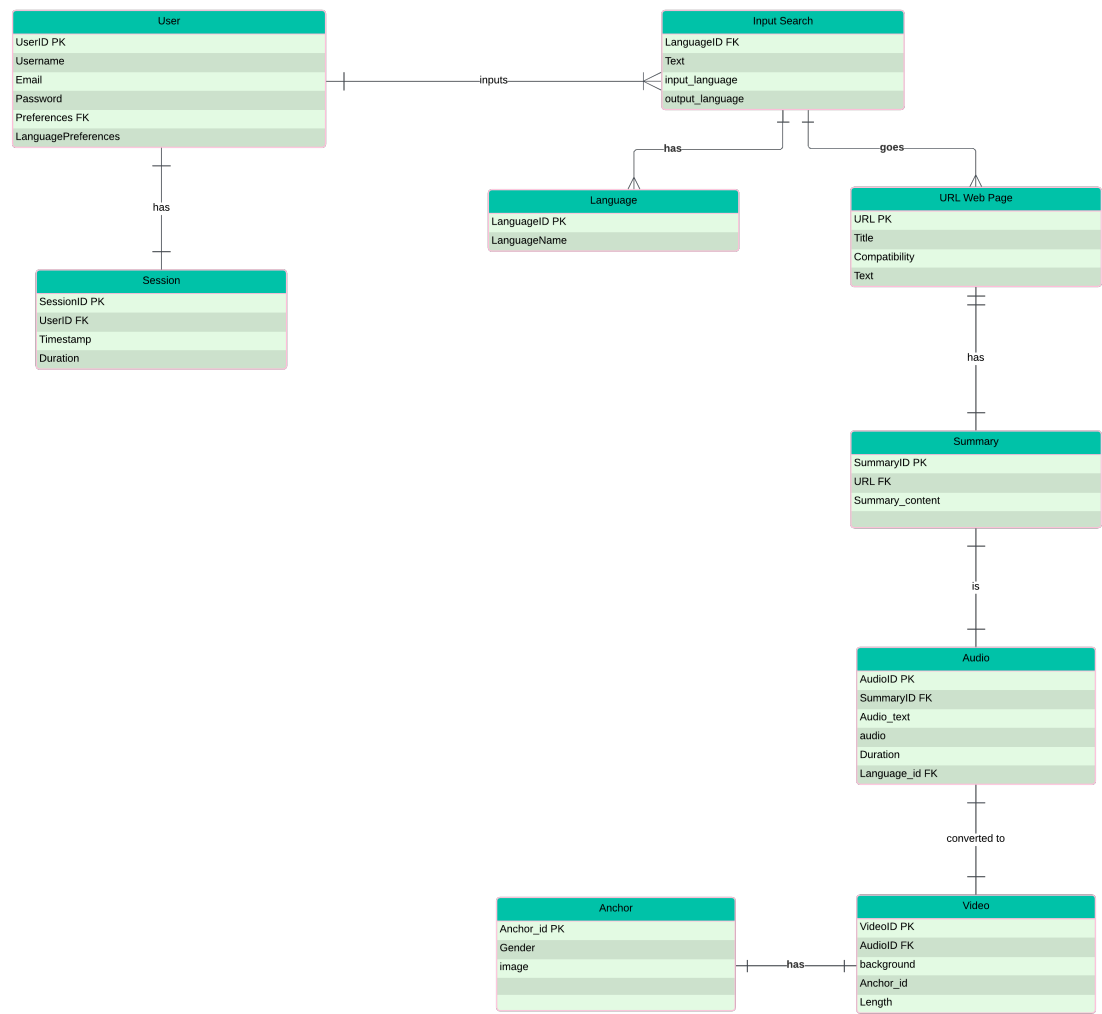


Figure 2.2: Domain Model Diagram

Chapter 3

System Overview

FUNCTIONALITY:

InfoMorph is a web-based platform that automates the conversion of any article or blog post into a concise and easily digestible summary, which is then narrated by an AI-generated video anchor. Users can either input a URL, and the platform scrapes the content, or they can input a topic, and the platform will provide a list of relevant URLs for the user to choose from. The selected content is then processed using natural language processing (NLP) to categorize and summarize the information. The summarized text is transformed into speech, and an AI video anchor narrates the content in a video format. These videos are designed to be short, engaging, and suitable for social media platforms like Instagram, making it easy for users to consume information quickly and efficiently.

CONTEXT: infoMorph addresses the broader issue of information overload and the growing preference for visual and summarized content especially among younger generations. Instead of focusing solely on news, the platform allows users to interact with a wide range of articles and blogs. With the shift towards digital consumption, many people find it challenging to sift through lengthy content, and InfoMorph provides a solution by presenting information in a simplified, engaging, and visually appealing way. This makes it suitable for anyone who wants to save time while staying informed, whether about current events, lifestyle, or any other subject.

DESIGN: InfoMorph is designed with simplicity and efficiency in mind. The platform uses web scraping tools to gather text from any URL, and NLP techniques categorize the content. Open-source large language models (LLMs) are employed to ensure the summaries are concise and meaningful. The summarized text is converted into speech using advanced text-to-speech technology, and a deepfake AI anchor narrates the content in a video format. The design ensures the final product is optimized for social media platforms, particularly for reels and short videos, ensuring the content is visually appealing and easily shareable.

3.1 Architectural Design

Our project, InfoMorph, is designed using N-Layered Architecture, in which, each layer handles a specific task. This architecture is made up of the user interface, where you interact with our application; the main body which are tasked with performing operations like summarizing articles and generating videos; the database, where we keep the text.

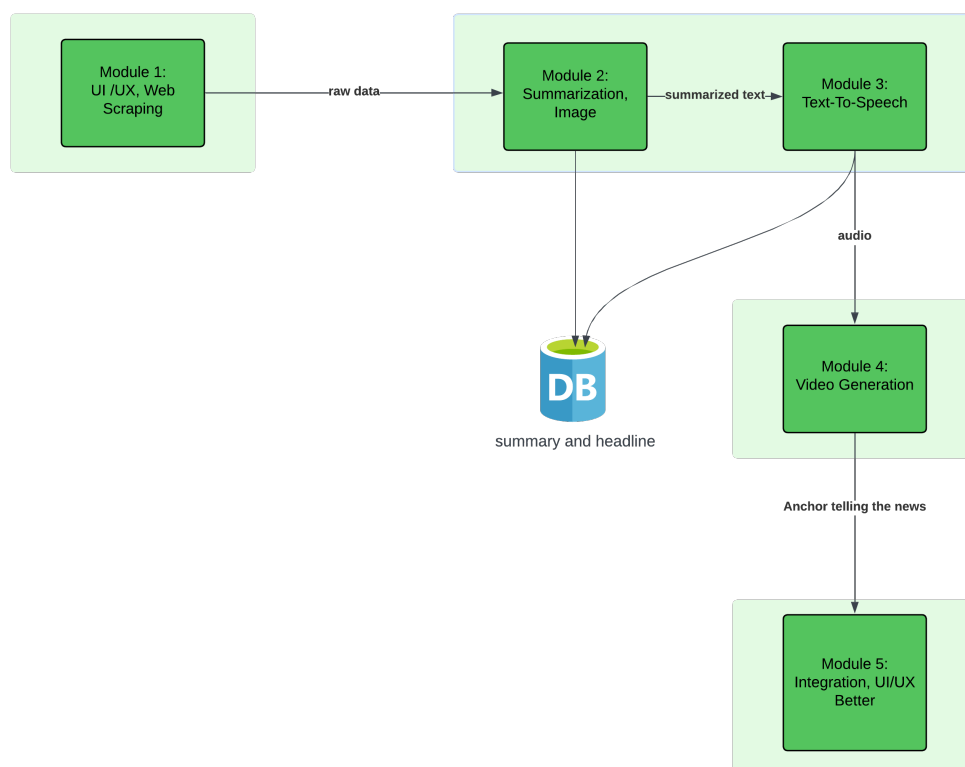


Figure 3.1: Box and Line Diagram

Our project is build on a simple foundation with five primary functions as listed in the diagram above. Each module will be designed individually with its function separate from the others. This approach will guarantee us a seamless and user-friendly experience for us as well as our potential users and will also make the system more manageable and scalable.

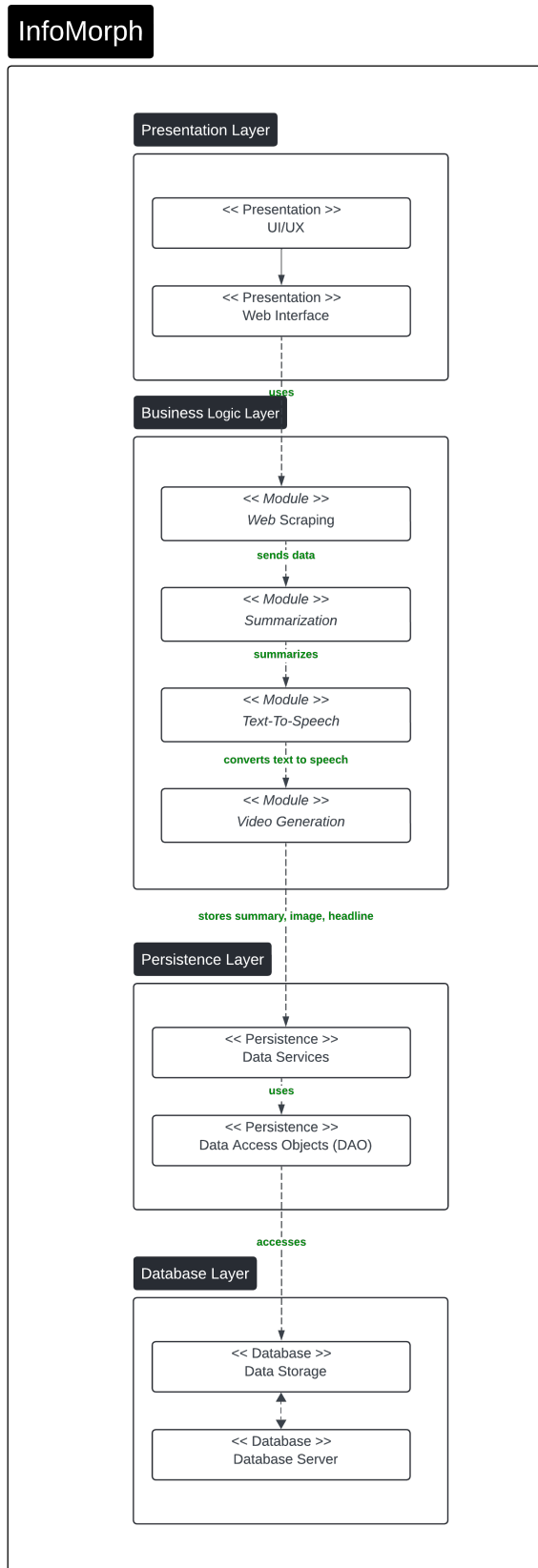


Figure 3.2: Architectural Diagram

The architectural diagram above tell us how we have used the N-Layered Architecture. The **Presentation Layer** manages all user interactions through the UI/UX components. The **Business Logic Layer** contains the most important modules i.e. Web Scrapping, Summarization, Text-To-Speech, and Video Generation. The **Persistence Layer** includes Data Access Objects(DAO) and Data Services that will help us maintain database transactions and data integrity. Lastly, is the **Database Layer** which will help us with maintaining our summarized text and image, that will be really helpful for us in terms of efficient data processing.

3.2 Design Models

Object Oriented Development Approach [2] Our project, InfoMorph uses this approach because it helps us facilitate our code better as it will allow us to organize our code better, improve maintainability and provide flexibility if we were to integrate new technologies in our code.

3.2.1 Select Topic

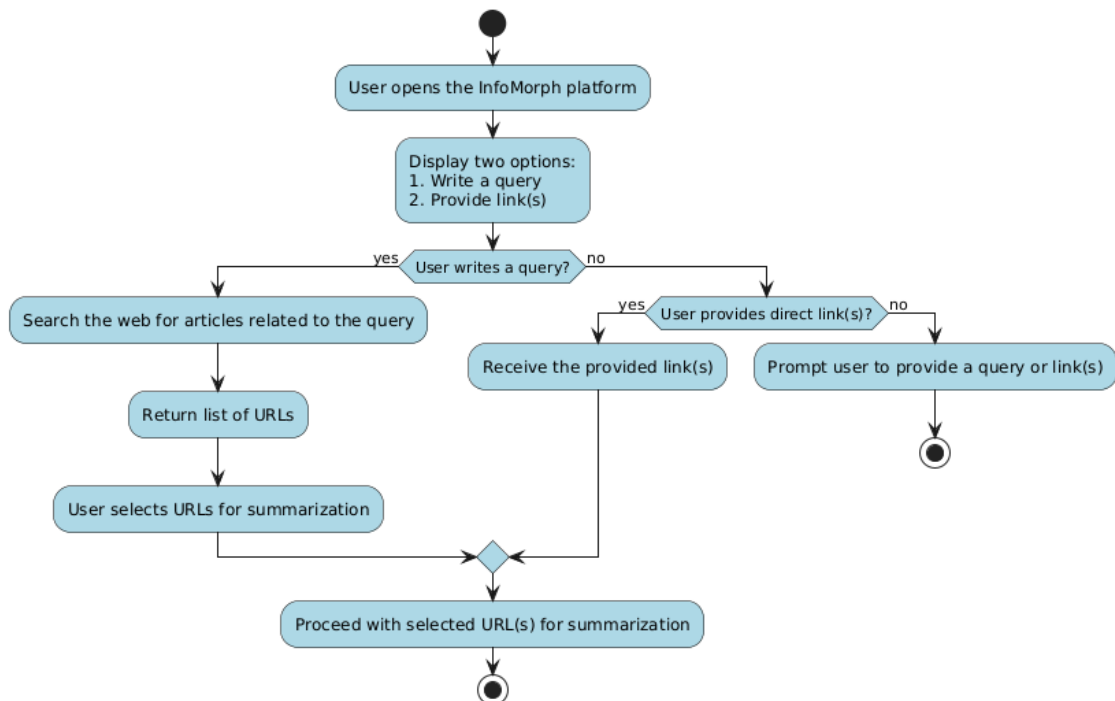


Figure 3.3: Select Topic Activity Diagram

3.2.2 Download

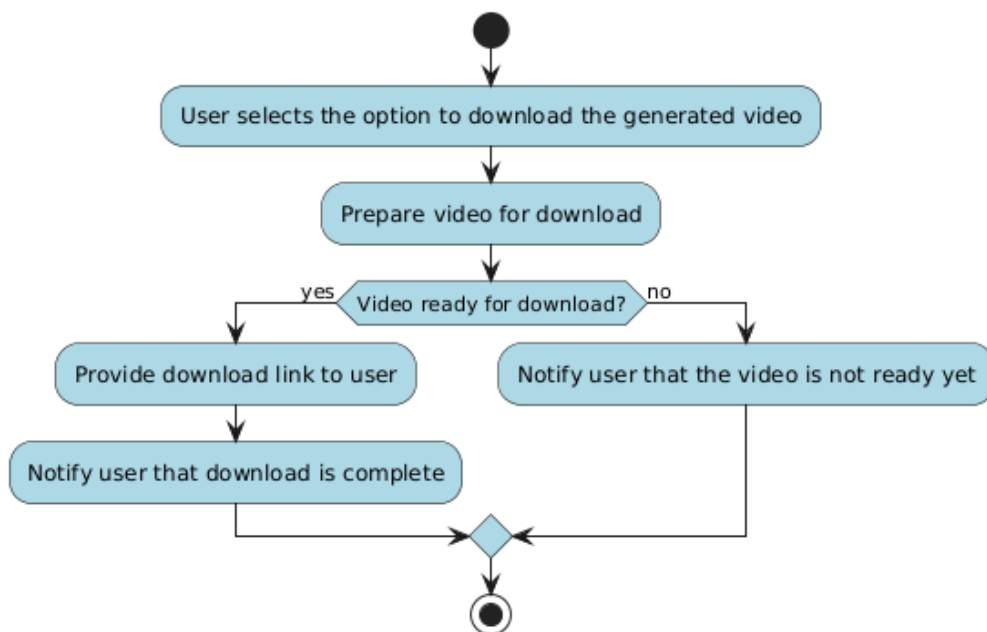


Figure 3.4: Download Activity Diagram

3.2.3 Feedback Handling

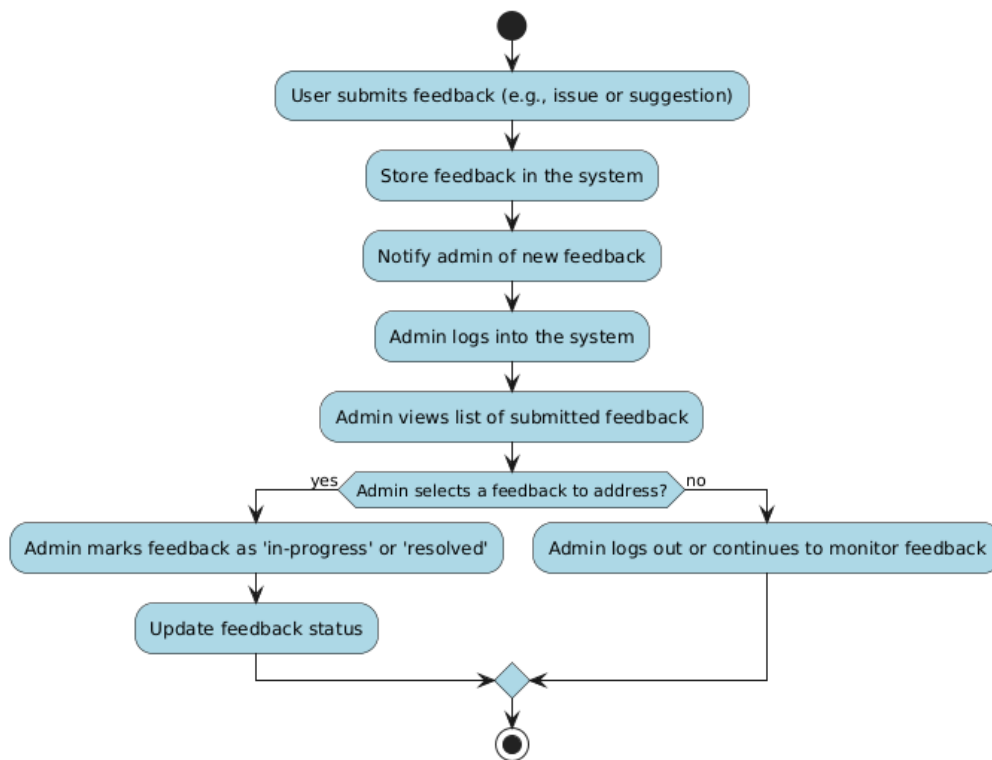


Figure 3.5: Feedback Handling Activity Diagram

3.2.4 Summarize Content

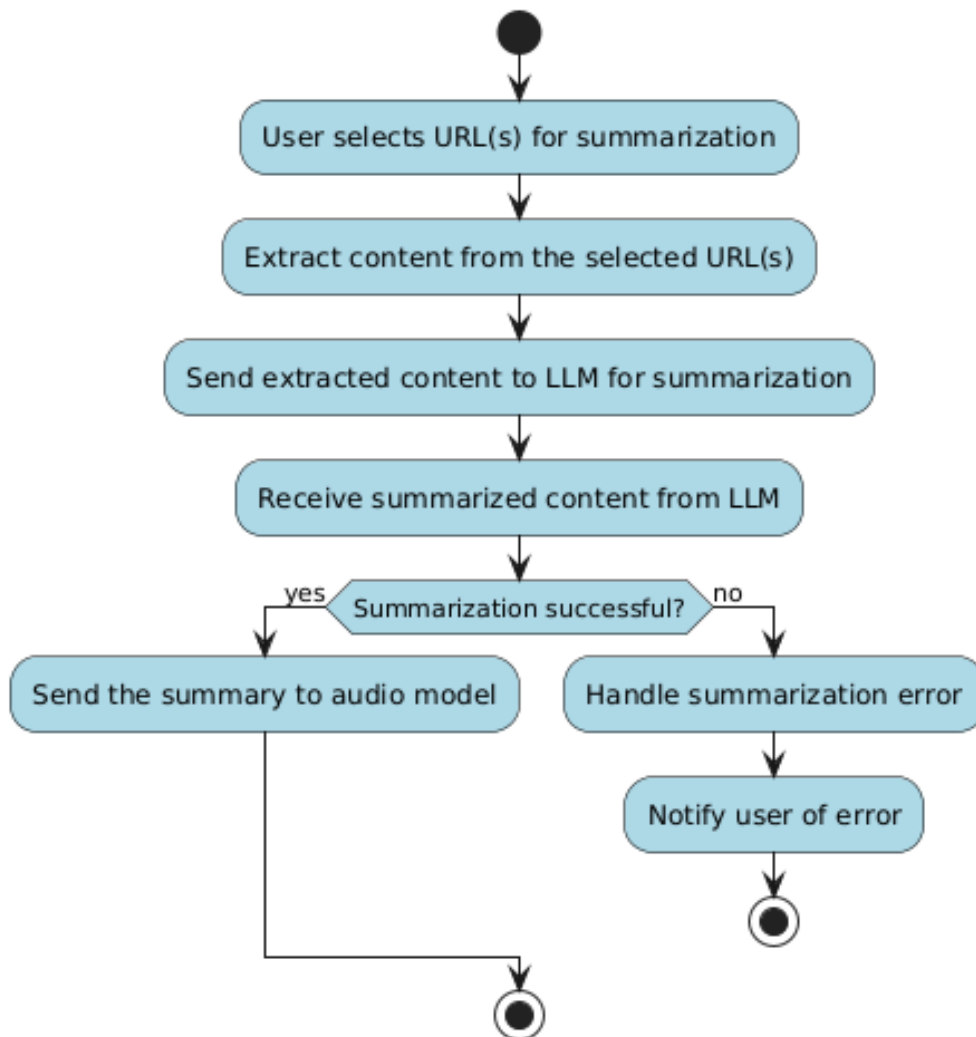


Figure 3.6: Summarize Content Activity Diagram

3.2.5 Text-To-Speech

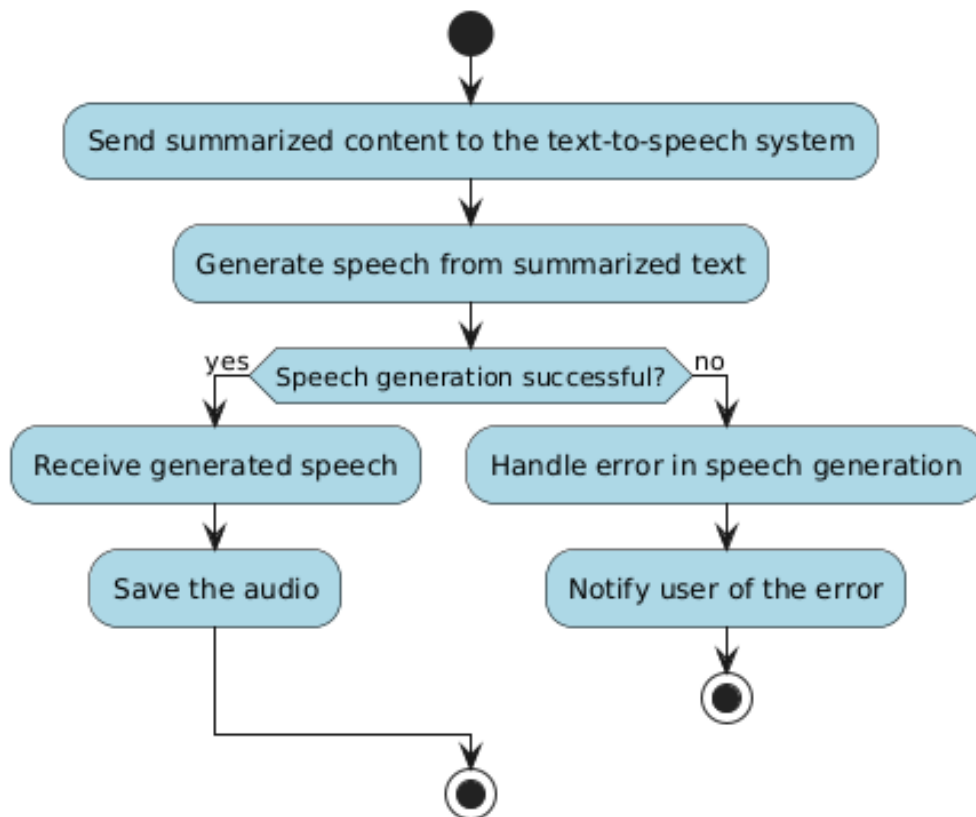


Figure 3.7: Text-To-Speech Activity Diagram

3.2.6 Video Generation

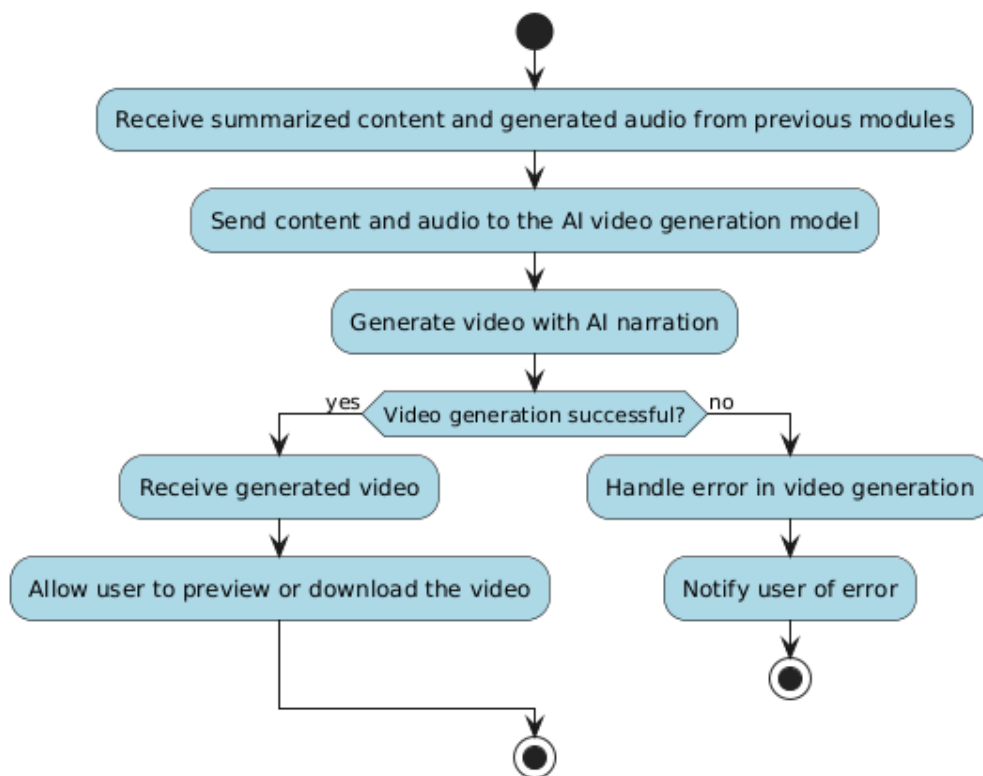


Figure 3.8: Video Generation Activity Diagram

3.2.7 Web Scraping

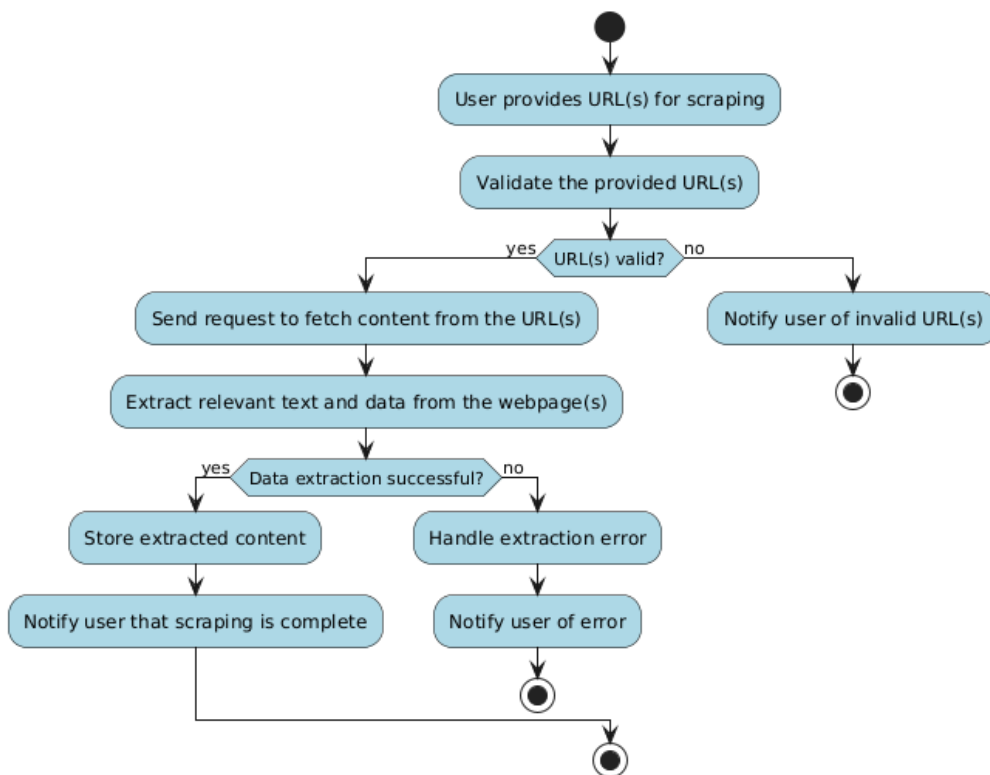


Figure 3.9: Web Scraping Activity Diagram

3.3 Class Diagram

3.3.1

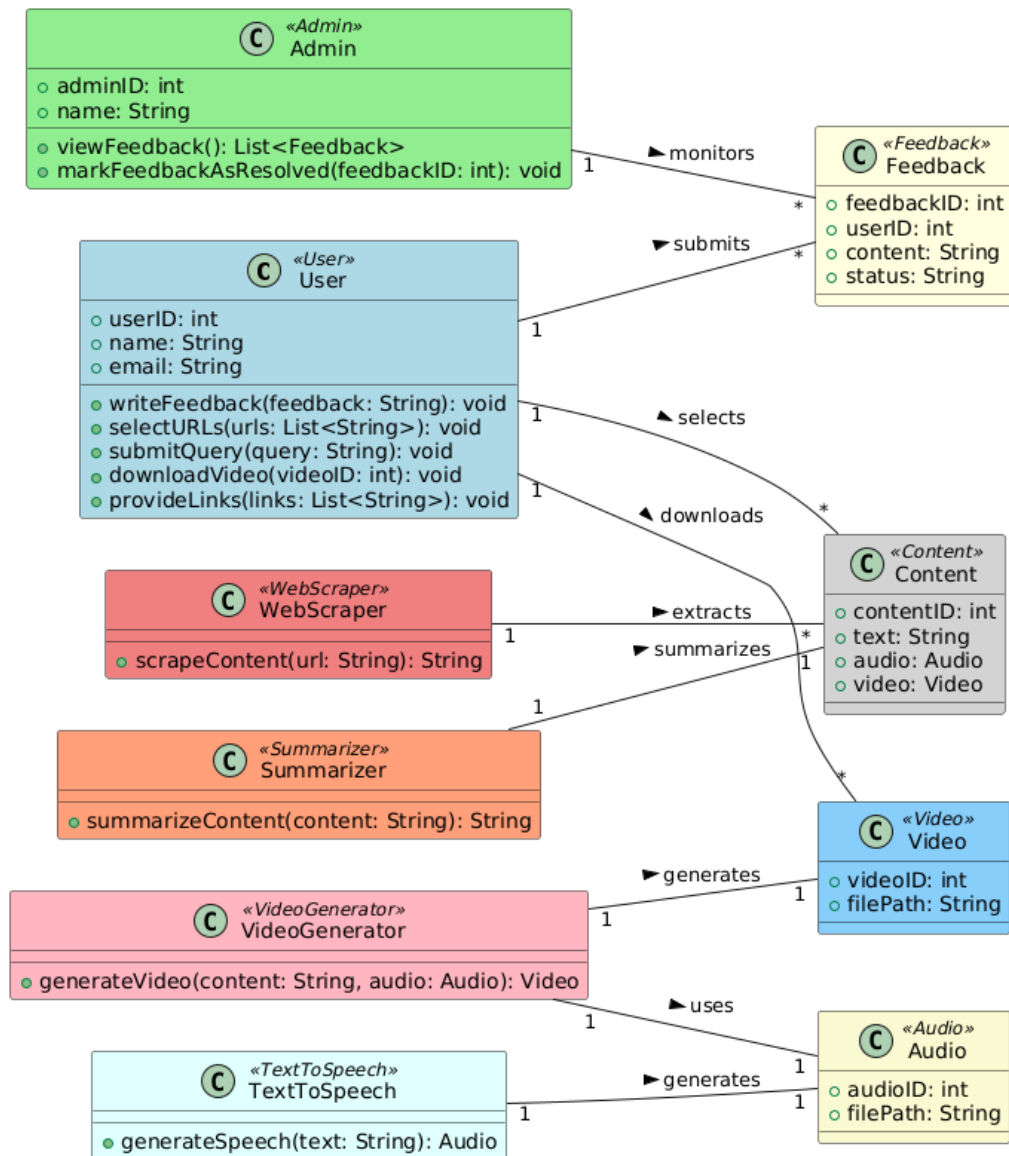


Figure 3.10: Class Diagram

3.4 Data Design

For our system, InfoMorph, it was very important to transform the information domain into data structures as it was linked to data handling, data storage and also data retrieval. InfoMorph deals with a range of data types like web content, user data, summarized text,

audios and video content. below is the explanation about how the information from this range of data is structured, then processed and finally organized in InfoMorph. [2]

3.4.1 Entity Identification

- **User:** It will manage user details and preferences, also will be connected to sessions and language.
- **Language:** It will be mainly 4 languages which user can choose to view from.
- **URL Webpage:** It will contain information regarding web pages used for scraping and summarization.
- **Audio:** It will contain the audio that was converted from text.
- **Anchor:** It will contain details about the AI News Anchors that will deliver the news or data.
- **Video:** It will link video to the corresponding audio and also will include the AI Anchor.

3.4.2 Data Storage

3.4.2.1 Database Management System(DBMS)

- **NoSQL DB:** We will be using MongoDB to store the summarized text along with its image.

3.4.2.2 File Storage

- **Media Files:** Audio and Video files will be either be stored in our own storage system or in a storage service on the cloud e.g Amazon S3.

3.4.2.3 Data Processing

- **Web Scraping:** We will be using web scraping to take content dynamically from the internet and process it.
- **Summarization:** We will be using an LLM Model to generate a high level summary from the scrapped web data.

- **Text-To-Speech:** We will be converting the generated summary from text to audio for our AI Anchor.
- **Video Generation:** This will be where the AI News Anchor will deliver the news.

3.4.3 Data Security and Integrity

Our system, InfoMorph will make sure our users data is safely stored and nothing happens to it. We will be doing this through:

- **Authorization:** According to their access, our audience will be able to view different parts of our project.
- **Backup:** Our system's data will have to be backup up regularly so that we don't lose any important data.

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